

Literature Survey

Real-Time UAV-Based Fire Detection and Intelligent Emergency Response Assistant: Fire Severity, Human Detection, Condition Assessment, and Rescue-Priority Estimation on Edge AI

1. Scope and Methodology

This survey covers 18 references spanning UAV-based fire/smoke detection, edge-AI deployment on embedded boards (Raspberry Pi / Jetson-class hardware), thermal and RGB human/victim detection, UAV-based casualty triage and pose/condition assessment, post-disaster survivor and damage-assessment datasets, and multi-task disaster-response learning. Sixteen references are drawn from peer-reviewed journals or juried venues (IEEE Access, IEEE Transactions on Vehicular Technology, Elsevier Information Fusion, Elsevier Fire Safety Journal, Springer Artificial Intelligence Review, Springer-published journals, Nature Scientific Data, MDPI Drones/Sensors/Mathematics/Journal of Imaging, and PLOS ONE), of which 15 were published in 2023-2026 (within the last three years). Two conference papers and one unreviewed preprint are included strictly as supporting material (shaded rows, tagged [C1], [C2], and [16]) and are not counted toward the 10-recent-reputed-venue requirement.

2. Comparative Table

Ref.	Year	Dataset	Method / Architecture	Key Metric & Value	Stated Limitation
[1]	2023	Custom UAV/machine-vision-course wildfire image set	Reduce-VGGNet (classification) + optimized CNN region detector	91.20% classification accuracy; 97.35% region-detection accuracy	Framework not validated on noisy, complex satellite imagery; limited to controlled machine-vision-course data
[2]	2024/2025	Literature corpus (pre-, active-, post-wildfire UAS studies)	Survey / taxonomy of AI-enabled UAV wildfire-management pipelines	N/A — qualitative survey covering 100+ primary studies	No unified benchmark or standard evaluation protocol exists across the surveyed UAS wildfire literature
[3]	2024	Custom fire dataset (7,187 images), field trials in Dhaka	YOLOv8n on Raspberry Pi 5 + Pi Camera Module 3, knowledge distillation	~8 FPS on-board; 38/65 correct real-world fire detections (58.5%)	Low real-world detection rate outside lab conditions; frame rate still modest for a nano board
[4]	2025	Kaggle forest-fire/smoke/non-fire image set	Lightweight MobileNetV2 CNN on Raspberry Pi 5	97.98% test accuracy; 0.77 s/frame latency (~1.3 FPS)	Sub-2-FPS throughput is too slow for fast-moving fire fronts; no on-board severity or victim analysis
[5]	2025	D-Fire / custom thermal-safety image set	YOLO11n detector + automated suppression trigger on Raspberry Pi	Real-time detection with integrated suppression actuation (reported precision/recall gains over YOLOv8-class baselines)	Suppression mechanism validated only in small-scale indoor/thermal-safety settings, not aerial/outdoor fire
[6]	2025	Literature corpus (UAV vision-based fire management)	Survey of detection, classification and segmentation DL techniques for UAV fire management	N/A — comparative survey of architectures across studies	Identifies absence of end-to-end systems that couple detection with downstream response/rescue tasks
[7]	2025	75,000+ image multi-source thermal benchmark (10 sources + new mountain set)	YOLOv8 (M/X), YOLOv9-E, YOLOv10 (M/X), RT-DETR-L, comparative embedded evaluation	mAP@0.5 up to 0.855 (YOLOv8); benchmarked across embedded platforms	Models trained on heterogeneous/urban data generalize poorly to unseen mountainous/forest terrain

Ref.	Year	Dataset	Method / Architecture	Key Metric & Value	Stated Limitation
[8]	2024	UAV-collected thermal video (custom)	YOLO detector + Kalman-filter multi-target tracking	Real-time survivor tracking with stable track association (frame-level qualitative results reported)	Thermal resolution loss and scene clutter degrade tracking in complex, multi-victim scenes
[9]	2025	C2A disaster dataset; COCO benchmark	LightSeek-YOLO (YOLOv11 + HGNetV2 backbone, Seek-DS, Seek-DH)	mAP@[0.5:0.95] = 0.473 (COCO); AP@small = 0.478 (C2A); 571.7 FPS (desktop GPU)	Real-time gains demonstrated on desktop GPU only; edge-board (e.g., Raspberry Pi) deployment not validated
[10]	2025	Maritime SAR aerial imagery	Enhanced YOLO11 (Space-to-Depth backbone module, content-aware neck upsampling, extra detection head)	Improved small-object precision/recall over baseline YOLO11 (maritime SAR benchmark)	Tuned specifically for open-water maritime scenes; not evaluated for land/fire/rubble environments
[11]	2025	Aerial small-object SAR image set	YOLO-based small-object detector for drone SAR	Improved small-object detection accuracy vs. baseline YOLO variants	Authors note a persistent gap between benchmark accuracy and real-field deployment reliability
[12]	2024	15 human subjects, 30 outdoor vital-sign trials	UAS-based respiratory-rate + body-pose triage classifier (camera-based)	89% triage classification accuracy; F1 = 0.94	Small pilot cohort (15 subjects); not tested under smoke, occlusion or multi-victim mass-casualty conditions
[13]	2024	Post-disaster UAV imagery (occluded scenes)	UGEN: CCGAN occlusion removal + ensemble (single-/multi-stage) detector over ORAN link	96.7% survivor-detection accuracy	Relies on ORAN network infrastructure for low-latency transfer; not designed for a fully offline, onboard pipeline
[14]	2023	RescueNet (Hurricane Michael UAV imagery)	Pixel-level semantic-segmentation damage-assessment dataset/benchmark	Dataset benchmark; class-wise mIoU reported for building/road/damage classes	Single-disaster-event imagery (hurricane); limited generalizability to fire-specific damage patterns
[15]	2024	Standard + UAV-specific low-resolution face datasets	UWS-YOLO: CNN-based low-resolution facial detector for UAV imagery	59.29% accuracy (vs. 27.43% RetinaFace, 46.59% YOLOv7); 11 ms inference	Accuracy remains modest in absolute terms; face-only cue is insufficient for full condition/priority assessment
[16]	2026*	Drone-captured casualty video + wearable sensor data (custom)	ATRACT: multi-modal (video pose/posture + physiological sensor) casualty-state classifier, CVAE augmentation	85.7% action-classification accuracy	Depends on body-worn sensors in addition to video, which may be unavailable for untagged disaster victims
[C1]	2019	Custom UAV forest-fire aerial images	Small-scale CNN via YOLOv3 on UAV platform	~83% recognition rate; 3.2 FPS	Low frame rate and accuracy by modern standards; superseded by later YOLO generations
[C2]	2023	Custom UAV thermal image set	YOLOv5 / YOLOv8 / SSD comparison for night-time thermal object detection	mAP@0.5 = 0.852 (YOLOv5), 0.855 (YOLOv8)	Evaluated on a comparatively small custom dataset; no on-board embedded deployment reported

Shaded rows = conference paper / unreviewed preprint cited as supporting material only (3 total, within the permitted limit). * ATRACT [16] is an arXiv preprint; its 2026 date reflects a training-data-cutoff-adjacent posting and is treated as supporting evidence, not as one of the 10 required recent peer-reviewed sources.

3. Research Gap

Read together, the comparative table exposes five gaps directly relevant to the proposed Raspberry Pi 4-based intelligent emergency-response assistant:

1. No end-to-end, single-platform pipeline. Every fire-detection study ([1]-[6], [C1]) and every human/victim-detection study ([7]-[11], [15], [C2]) addresses one subtask in isolation. None of the surveyed works integrates fire detection, fire-severity estimation, human detection, condition assessment, rescue-priority ranking, and geotagged mission-report generation into a single onboard pipeline.
2. Absence of jointly-annotated benchmark data. Fire datasets (FASDD, D-Fire, Kaggle wildfire sets used in [1], [3]-[5]) and victim/casualty datasets (C2A, HERIDAL-type sets, RescueNet [14]) are collected and annotated independently, so no public dataset links fire severity, victim location, and victim condition within the same aerial scene — a prerequisite for training a unified severity-to-priority model.
3. Triage and condition-assessment methods are unvalidated under fire/smoke degradation. The most capable condition-assessment systems ([12], [13], [16]) were evaluated in clear-weather mass-casualty or occlusion scenarios, not under smoke, heat-shimmer, or fire-lit conditions that a fire-response UAV will actually encounter.
4. Edge-AI demonstrations skip the lowest-cost hardware tier. Real-time embedded fire detection has been shown on Raspberry Pi 5 ([3], [4], [5]) and on Jetson/GPU-class boards ([7], [9] desktop GPU only); no reviewed study demonstrates the combined fire + human + condition + priority pipeline running in real time on the more constrained, lower-cost Raspberry Pi 4 targeted by this project.
5. No automated fusion of severity with rescue-priority decision output. Detection-and-communication frameworks such as UGEN [13] and multi-task disaster classifiers optimize detection accuracy and network latency, but none of the surveyed works closes the loop by converting detected fire severity and victim condition into an automatically ranked rescue-priority list and a structured mission report for first responders.

4. Reference List

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